

goxpyriment — API Reference

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goxpyriment API Reference

This guide documents the complete public API of the goxpyriment framework, organized by package.

Note that you can access the source code documentation by running pkgsite and opening <http://localhost:8080>. To install pkgsite:

```
go install golang.org/x/pkgsite/cmd/pkgsite@latest
```

See [Viewing the documentation locally](#) for details (and for serving this site offline with zensical).

Package Overview

control/	← experiment lifecycle and orchestration (start here)
stimuli/	← visual and audio stimulus objects
media/	← multi-movie playback with hardware-verified display onsets
apparatus/	← SDL window/renderer, keyboard, mouse, gamepad, gamma corrector
results/	← experiment data file and output file
design/	← trial/block structure and randomization
clock/	← timing utilities
geometry/	← coordinate conversion helpers

triggers/ ← hardware trigger devices (EEG sync, etc.)

Package control

Import: `github.com/chrplr/goxpyriment/control`

Boilerplate

Every experiment starts the same way:

```
exp := control.NewExperimentFromFlags("My Experiment", control.Black, control.White, 32)
defer exp.End()

err := exp.Run(func() error {
    // trial loop
    return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
```

Pre-experiment Setup Dialog

`GetParticipantInfo` opens a graphical SDL window **before** the experiment starts to collect participant demographics, monitor properties, and display preferences. Call it before `NewExperiment` / `NewExperimentFromFlags`.

```
fields := append(control.StandardFields, control.FullscreenField)
info, err := control.GetParticipantInfo("My Experiment", fields)
if err != nil {
    log.Fatalf("setup cancelled: %v", err) // user pressed Escape or closed window
}
```

```
// FieldType selects how a field is rendered.
type FieldType int
const (
    FieldText      FieldType = iota // text input box (default)
    FieldCheckbox                // tick-box; value is "true" or "false"
)

// InfoField describes one entry in the dialog.
type InfoField struct {
    Name      string // key in the returned map
    Label     string // displayed label
    Default   string // initial value
}
```

```

    Type      FieldType // FieldText (default) or FieldCheckbox
}

```

Types

Pre-built field sets

Variable	Fields
control.ParticipantFields	subject_id, age, gender, handedness
control.MonitorFields	screen_width_cm, viewing_distance_cm, refresh_rate_hz
control.StandardFields	ParticipantFields + MonitorFields
control.FullscreenField	Checkbox: fullscreen ("true" / "false")

Function

Function	Description
GetParticipantInfo(title string, fields []InfoField) (map[string]string, error)	Shows the dialog and returns collected values. Returns ErrCancelled if the user closes or presses Escape without confirming.

Session persistence All values except subject_id are saved to ~/.cache/goxpyriment/last_session on OK and pre-filled on the next run. subject_id is always reset.

```

info, err := control.GetParticipantInfo("My Experiment", fields)
// ...
fullscreen := info["fullscreen"] == "true"
width, height := 0, 0
if !fullscreen {
    width, height = 1024, 768
}
exp := control.NewExperiment("My Experiment", width, height, fullscreen,
    control.Black, control.White, 32)

// Set Info (and SubjectID) BEFORE Initialize – they are written to the info file autom
exp.SubjectID, _ = strconv.Atoi(info["subject_id"])
exp.Info = info

if err := exp.Initialize(); err != nil { log.Fatal(err) }
defer exp.End()

```

Using the fullscreen checkbox and persisting to the data file Initialize() writes a --PARTICIPANT INFO block to the companion -info.txt file whenever exp.Info is non-nil at that point. No explicit call to WriteParticipantInfo is needed.

```
control.ErrCancelled // returned when the user cancels the dialog
```

Sentinel error

Constructor Functions

Function	Description
<code>NewExperimentFromFlags(name string, bg, fg Color, fontSize float32) *Experiment</code>	Creates and fully initializes an experiment from -w (windowed 1024×768), -d N (display index, -1 = primary), and -s N (subject ID) command-line flags. Calls <code>log.Fatal</code> on error. This is the preferred entry point.
<code>NewExperiment(name string, width, height int, fullscreen bool, bg, fg Color, fontSize float32) *Experiment</code>	Lower-level constructor; call <code>Initialize()</code> before use.

Lifecycle Methods

Method	Description
<code>exp.Initialize() error</code>	Initializes SDL, audio, window, renderer, font, and data file.
<code>exp.End()</code>	Cleans up all resources. Always defer <code>exp.End()</code> immediately after construction.
<code>exp.Run(logic func() error) error</code>	Runs the main trial loop on the SDL main thread. Return <code>control.EndLoop</code> to exit cleanly.
<code>exp.HideCursor() error</code>	Hides the mouse cursor. Call after <code>Initialize()</code> to prevent the cursor from appearing over stimuli.
<code>exp.ShowCursor() error</code>	Makes the mouse cursor visible again.

Presentation Methods

Method	Description
<code>exp.Show(stim VisualStimulus) error</code>	Clear → draw → flip. The standard one-call stimulus presentation.

Method	Description
<code>exp.ShowTS(stim VisualStimulus) (uint64, error)</code>	Clear → draw → flip, and return the SDL nanosecond timestamp captured immediately after the VSYNC flip. Use with <code>GetKeyEventTS</code> for hardware-precision RT measurement.
<code>exp.ShowTimed(stim VisualStimulus, durationMs int) error</code>	<code>Show(stim) + Wait(durationMs)</code> in one call. For fixation crosses, cues, and passive stimulus viewing.
<code>exp.ShowAndGetRT(stim VisualStimulus, keys []Keycode, timeoutMs int) (Keycode, int64, error)</code>	Clears stale keyboard events, shows stim with hardware-precise onset timing, waits for a key, and returns (key, rtMs, error). Pass <code>timeoutMs = -1</code> for no timeout; returns (0, 0, nil) on timeout. This is the canonical single-stimulus RT measurement call.
<code>exp.ShowEndMessage(message string) error</code>	Display a centered completion message and wait for any key. For end-of-experiment screens.
<code>exp.ShowInstructions(text string) error</code>	Display centered text and wait for spacebar.
<code>exp.Blank(ms int) error</code>	Clear and flip screen, then wait ms milliseconds.
<code>exp.Wait(ms int) error</code>	Wait ms ms while pumping SDL events (ESC-abortable).
<code>exp.ShowSplash(waitForKey bool) error</code> <code>exp.Flip() error</code>	Show experiment name + version splash. Present the backbuffer (VSYNC-locked when VSync is enabled).

Input

Method	Description
<code>exp.Keyboard</code>	* <code>apparatus.Keyboard</code> — see Keyboard section
<code>exp.Mouse</code> <code>exp.PollEvents(handle func(sdl.Event) bool) EventState</code>	* <code>apparatus.Mouse</code> — see Mouse section Process all pending SDL events; optionally forward to a handler. Returns <code>EventState</code> including nanosecond timestamps.
<code>exp.HandleEvents() (Keycode, uint32, error)</code>	Convenience wrapper: returns (key, mouseButton, error).

`EventState` now includes SDL event timestamps:

```

type EventState struct {
    LastKey          sdl.Keycode
    LastMouseButton  uint32
    LastKeyTimestamp uint64  // SDL nanosecond timestamp of the last key event
    LastMouseTimestamp uint64 // SDL nanosecond timestamp of the last mouse event
    QuitRequested    bool
}

```

Design and Data

Method	Description
<code>exp.AddDataVariableNames(names []string)</code>	Register CSV column names for the data file.
<code>exp.Data.Add(values ...interface{})</code>	Append a data row. Subject ID is prepended automatically.
<code>exp.AddBlock(b *design.Block, copies int)</code>	Add trial blocks to the experiment.
<code>exp.ShuffleBlocks()</code>	Randomize block presentation order.
<code>exp.AddBWSFactor(name string, conditions []interface{})</code>	Register a between-subjects factor for Latin-square counterbalancing.
<code>exp.GetPermutedBWSFactorCondition(name string) interface{}</code>	Return this subject's condition for a BWS factor.
<code>exp.Design</code>	*design.Experiment — full design object
<code>exp.Info</code>	map[string]string — values from GetParticipantInfo; set before Initialize() to persist them automatically to the -info.txt file

Font and Display

Method	Description
<code>exp.LoadFont(path string, size float32) error</code>	Load a TTF font from file.
<code>exp.LoadFontFromMemory(data []byte, size float32) error</code>	Load a TTF font from a byte slice.
<code>exp.SetVSync(vsync int) error</code>	Toggle vertical sync (1 = on, 0 = off).
<code>exp.SetLogicalSize(w, h int32) error</code>	Set device-independent logical resolution.
<code>exp.SetOutputDirectory(dir string)</code>	Override default data file directory (~/.goxpy_data).

Gamma Correction

Standard monitors apply a power-law transfer function $L(V) = k \cdot (V/255)^\gamma$ ($\gamma \approx 2.2$ for sRGB displays). Equal steps in RGB values do **not** produce equal steps in physical

luminance. Use SetGamma to enable inverse-gamma correction.

Method	Description
exp.SetGamma(gamma float64)	Install a uniform inverse-gamma corrector. Call once after Initialize().
exp.CorrectColor(c sdl.Color) sdl.Color	Apply gamma correction to a color. Returns c unchanged when no corrector is set.
exp.GammaCorrector	*apparatus.GammaCorrector — set directly for per-channel calibration.

```
// Uniform gamma (typical sRGB monitor)
exp.SetGamma(2.2)

// Per-channel gamma (from photometer measurements)
exp.GammaCorrector = apparatus.NewGammaCorrector(2.1, 2.2, 2.3)

// Use in trial loop – specify colors in linear luminance space (0–255)
disk := stimuli.NewFilledCircle(exp.CorrectColor(control.RGB(128, 128, 128)), radius)
```

The apparatus.GammaCorrector type is also available directly:

```
gc := apparatus.NewGammaCorrectorUniform(2.2)
corrected := gc.CorrectColor(sdl.Color{R: 128, G: 128, B: 128, A: 255})
// corrected.R ≈ 186 – the physical digital value for 50% luminance on  $\gamma=2.2$ 
```

Colors, Types, and Constants

```
// Named colors
control.Black, White, Red, Green, Blue, Yellow, Magenta, Cyan
control.Gray, DarkGray, LightGray

// Type aliases (so you only need to import "control")
type Color    = sdl.Color
type Keycode  = sdl.Keycode
type FPoint   = sdl.FPoint
type FRect    = sdl.FRect

// Constructors
control.RGB(r, g, b uint8) Color
control.RGBA(r, g, b, a uint8) Color
control.Point(x, y float32) FPoint
control.Origin() FPoint // returns (0, 0)

// Font helpers
control.FontFromFile(path string, size float32) (*ttf.Font, error)
```

```

control.FontFromMemory(data []byte, size float32) (*ttf.Font, error)

// Loop control
control.EndLoop           // sentinel error: return from Run callback to exit
control.IsEndLoop(err)    // test whether an error is the EndLoop sentinel

// Keyboard codes
control.K_SPACE, K_ESCAPE, K_RETURN, K_BACKSPACE, K_TAB
control.K_UP, K_DOWN, K_LEFT, K_RIGHT
control.K_A ... K_Z           // full alphabet
control.K_0 ... K_9           // digit row
control.K_KP_0 ... K_KP_9, K_KP_ENTER, K_KP_PLUS, K_KP_MINUS // numeric keypad
control.K_MINUS, K_PLUS, K_EQUALS, K_LEFTBRACKET, K_RIGHTBRACKET // punctuation

// Mouse buttons
control.BUTTON_LEFT, BUTTON_RIGHT

```

Tip: The full alphabet, digit row, keypad, and common punctuation keys are re-exported, so experiment code never needs to import `go-sdl3` just to name a key. For a rarely-used code not listed in `defaults.go`, import `go-sdl3/sdl` directly and use `sdl.K_...`

Audio

```

exp.AudioDevice // sdl.AudioDeviceID – pass to Sound.PreloadDevice()

// Top-level helper (call before NewExperiment)
control.SetAudioSampleFrames(frames int) // set audio buffer size (256–2048)

```

Microphone

```

exp.Microphone // *apparatus.Microphone – nil until OpenMicrophone is called

// Open the default recording device (nil = F32LE mono 44100 Hz)
exp.OpenMicrophone(spec *sdl.AudioSpec) error

// Closed automatically by exp.End()

```

Package stimuli

Import: `github.com/chrplr/goxpyriment/stimuli`

Interfaces

```
type Stimulus interface {
    Present(screen *apparatus.Screen, clear, update bool) error
    Preload() error
    Unload() error
}

type VisualStimulus interface {
    Stimulus
    Draw(screen *apparatus.Screen) error
    GetPosition() sdl.FPoint
    SetPosition(pos sdl.FPoint)
}
```

GPU textures are **lazily allocated** on the first Draw. To force early allocation (for timing-sensitive code), use:

```
stimuli.PreloadVisualOnScreen(screen, stim)    // single stimulus
stimuli.PreloadAllVisual(screen, []VisualStimulus{...}) // slice
```

Visual Stimuli

Text

Constructor	Description
<code>NewTextLine(text string, x, y float32, color Color) *TextLine</code>	Single line of text.
<code>NewTextBox(text string, width int32, pos FPoint, color Color) *TextBox</code>	Word-wrapped multi-line text.

Both support a `Font *ttf.Font` field — set it to override the screen default.

Shapes

Constructor	Description
<code>NewFixCross(size, lineWidth float32, color Color) *FixCross</code>	Fixation cross centered at (0, 0).
<code>NewCircle(radius float32, color Color) *Circle</code>	Filled circle.
<code>NewRectangle(cx, cy, w, h float32, color Color) *Rectangle</code>	Filled rectangle.
<code>NewLine(x1, y1, x2, y2 float32, color Color) *Line</code>	Line segment.

Images and Video

Constructor/Function	Description
NewPicture(filePath string, x, y float32) *Picture	Image loaded from file (PNG, JPG, BMP...).
NewPictureFromMemory(data []byte, x, y float32) *Picture	Image loaded from embedded bytes.
PlayGv(screen, path string, x, y float32) ([]UserEvent, error)	Play a .gv (LZ4-compressed RGBA) video file, VSYNC-locked.
NewGvVideo(path string) (*GvVideo, error)	Open a .gv file for frame-by-frame access.

Psychophysics Stimuli

Constructor	Description
NewGaborPatch(sigma, theta, lambda, phase, psi, gamma float64, bgColor Color, size float32) *GaborPatch	Static Gabor patch. theta in degrees, lambda = spatial wavelength in pixels.
NewDotCloud(radius float32, bgColor, dotColor Color) *DotCloud	Static random-dot cloud. Call Make(nDots, dotRadius, gap) to populate.
NewRDS(imgSize, innerSize [2]int, shift, gap, scale int) *RDS	Random-dot stereogram (side-by-side pair).
NewVisualMask(w, h, dotW, dotH float32, bgColor, dotColor Color, pct int) *VisualMask	Random-dot masking stimulus. pct = dot fill percentage 0-100.

Composite / Interactive

Constructor	Description
NewThermometerDisplay(size FPoint, nSegments int, state, goal float32) *ThermometerDisplay	Segmented progress bar. State and Goal in 0-100.
NewChoiceGrid(choices []string, maxSelect int, prompt string) *ChoiceGrid	Multiple-choice button grid (mouse + keyboard). See below.
NewTextInput(msg string, pos FPoint, boxW float32, bgColor, frameColor, textColor Color) *TextInput	Free-text keyboard input box. Call ti.Get(screen, keyboard).
NewMenu(items []string) *Menu	Numbered keyboard-navigable list. Call m.Get(screen, keyboard, initialSel).

ChoiceGrid

```
cg := stimuli.NewChoiceGrid(choices, maxSelect, prompt)
cg.Cols = 7           // optional: set column count (0 = auto)

selections, err := cg.Get(exp.Screen, exp.Keyboard)
// selections is a []string preserving selection order
```

- MaxSelect > 0: auto-submits after N selections.
- MaxSelect == 0: participant presses ENTER or SPACE to submit.
- BACKSPACE removes the last selection.
- Both mouse click and matching keypress (single-char labels) activate buttons.

Menu

```
m := stimuli.NewMenu([]string{"Option A", "Option B", "Option C"})
m.Pos = sdl.FPoint{X: 0, Y: 0} // optional: reposition (default = screen center)
m.HighlightColor = control.Yellow // optional: override highlight color

idx, err := m.Get(exp.Screen, exp.Keyboard, 0) // 0 = initially highlight first item
// idx is 0-based; -1 + sdl.EndLoop on ESC/quit
```

Navigation: UP/DOWN arrows move the highlight; ENTER or SPACE confirms; number keys 1-9 (0 for tenth) select and confirm directly. The selected item is shown in HighlightColor with a > prefix; others use TextColor. LineSpacing controls vertical item spacing (0 = auto from font height).

Animated / VSYNC-locked Loops

All three functions disable GC, lock to VSYNC, and return (MotionResult, error).

```
type MotionResult struct {
    Key      sdl.Keycode // interrupt key pressed (0 if none)
    Button   uint8      // mouse button pressed (0 if none)
    RTms     int64      // ms from first frame to response (or total duration on timeout)
}
```

Function	Description
PresentMovingDotCloud(screen, nDots int, dotRadius, cloudRadius float32, center FPoint, speedPxPerSec float32, maxDurationMs int64, interruptKeys []Keycode, catchMouse bool, dotColor, bgColor Color) (MotionResult, error)	Animated random-dot cloud. Each dot moves at a fixed speed and respawns when it exits the boundary.

Function	Description
PresentMovingGrating(screen, width, height float32, center FPoint, orientation, spatialFreq, temporalFreq, contrast, bgLuminance float64, maxDurationMs int64, interruptKeys []Keycode, catchMouse bool) (MotionResult, error)	Drifting sinusoidal grating in a rectangular aperture.
PresentMovingGabor(screen, size float32, sigma float64, center FPoint, orientation, spatialFreq, temporalFreq, contrast, bgLuminance float64, maxDurationMs int64, interruptKeys []Keycode, catchMouse bool) (MotionResult, error)	Drifting Gabor patch with Gaussian envelope (alpha-blended edges).

Spatial frequency is in **cycles per pixel** (e.g. 0.05 = one cycle every 20 px). Temporal frequency is in **Hz**. Orientation is in **degrees from horizontal** (0° = vertical bars drifting right).

Stimulus Streams (High-Precision RSVP)

Stream functions disable GC, lock every onset and offset to a VSYNC boundary, and return ([]UserEvent, []TimingLog, error).

```

type UserEvent struct {
    Event      sdl.Event      // raw SDL event (KeyboardEvent, MouseButtonEvent, ...)
    Timestamp   time.Duration  // time relative to stream start (Go clock, ms precision)
    TimestampNS uint64         // SDL3 hardware event timestamp, nanoseconds (same clock)
}

type TimingLog struct {
    Index        int
    TargetOn     time.Duration
    ActualOnset  time.Duration // DIAGNOSTIC (Go clock): first-frame draw, stream-relat
    ActualOffset time.Duration // DIAGNOSTIC (Go clock): after last on-frame, stream-re
    OnsetNS      uint64         // AUTHORITATIVE: SDL3 ns timestamp (sdl.TicksNS clock)
    OffsetNS     uint64         // AUTHORITATIVE: SDL3 ns timestamp (sdl.TicksNS clock)
}

```

The OnsetNS/OffsetNS pair is the authoritative timing record: both are on the SDL3 nanosecond clock (sdl.TicksNS()) — the same clock that stamps input events (UserEvent.TimestampNS, Keyboard.GetKeyEventTS) and VSYNC flips (Screen.FlipTS). A reaction time or a displayed duration is a plain subtraction within this one clock: int64(event.TimestampNS - l.OnsetNS), or int64(l.OffsetNS - l.OnsetNS).

ActualOnset/ActualOffset are Go-clock (time.Now) **diagnostics only**, kept for coarse pacing checks. They live on a different timebase (different origin) from the NS fields

and from input events, so they must never be subtracted from an event timestamp or logged as a canonical onset/offset. Use the NS fields for any real timing.

OnsetNS is zero only when a stimulus was never displayed (zero on-frames); OffsetNS is the VSYNC timestamp of the flip that takes the stimulus down, and for the final element of a contiguous stream (which has no ISI flip of its own) it is synthesised from the onset plus the on-frame count so that it, too, is on the SDL clock rather than one frame early.

UserEvent.TimestampNS and TimingLog.OnsetNS/OffsetNS are all on the SDL3 nanosecond clock, so reaction times measured during a stream can be computed with full hardware precision:

```
for _, ev := range events {
    if ev.Event.Type == sdl.EVENT_KEY_DOWN {
        // Find the stimulus that was on-screen when the key was pressed
        for _, l := range logs {
            if ev.TimestampNS >= l.OnsetNS && ev.TimestampNS < l.OffsetNS {
                rtNS := int64(ev.TimestampNS - l.OnsetNS)
                fmt.Printf("RT from stimulus %d: %d ms\n", l.Index, rtNS/1_000_000)
            }
        }
    }
}
```

Searching event lists

Function	Description
FirstKeyPress(events []UserEvent, key sdl.Keycode) (UserEvent, bool)	Returns the first KEY_DOWN event matching key from the slice, plus a found flag.

```
if ev, ok := stimuli.FirstKeyPress(events, sdl.K_SPACE); ok {
    fmt.Printf("Space pressed at %d ms\n", ev.Timestamp.Milliseconds())
}
```

```
// RSVP text stream – simplest entry point
events, logs, err := stimuli.PresentStreamOfText(
    exp.Screen, words, durationOn, durationOff, x, y, color,
)

// Image/mixed stream
elements := stimuli.MakeRegularVisualStream(stims, durationOn, durationOff)
events, logs, err := stimuli.PresentStreamOfImages(exp.Screen, elements, x, y)

// Irregular timing
```

```
elements, err := stimuli.MakeVisualStream(stims, onsetMs, durationMs)
events, logs, err := stimuli.PresentStreamOfImages(exp.Screen, elements, x, y)
```

Visual Streams

```
// Regular timing
elements := stimuli.MakeRegularSoundStream(sounds, durationOn, durationOff)
events, logs, err := stimuli.PlayStreamOfSounds(elements)

// Irregular timing
elements, err := stimuli.MakeSoundStream(sounds, onsetMs, durationMs)
events, logs, err := stimuli.PlayStreamOfSounds(elements)
```

Audio Streams `sounds` is `[]stimuli.AudioPlayable` — satisfied by both `*Sound` and `*Tone`. The returned `TimingLog` carries `OnsetNS/OffsetNS` on the SDL3 clock (captured at the `Play()` instant and the end of the on-phase), so audio-stream onsets are directly comparable with input-event timestamps, same as visual streams.

Mixed Streams `PresentStreamOfStimuli` presents a heterogeneous, **sequential** stream that freely mixes visual and audio stimulus types (anything satisfying the broad `Stimulus` interface). It uses the same VSYNC-locked, GC-disabled loop as `PresentStreamOfImages` (which now delegates to it).

```
elements := []stimuli.StreamElement{
    {Stimulus: stimuli.NewTextLine("Ready?", 0, 0, color), DurationOn: 600 * time.Millisecond,
    {Stimulus: pic, DurationOn: 800 * time.Millisecond}, // held while the tone plays
    {Stimulus: tone, DurationOn: 400 * time.Millisecond, DurationOff: 200 * time.Millisecond}
}
events, logs, err := stimuli.PresentStreamOfStimuli(exp.Screen, elements, x, y)

// Builders mirror the visual/sound ones but take []stimuli.Stimulus:
elements = stimuli.MakeRegularStream(stims, durationOn, durationOff)
elements, err = stimuli.MakeStream(stims, onsetMs, durationMs)
```

Per-element semantics:

- **Visual** elements (those satisfying `VisualStimulus`) are centered on `(x, y)` and redrawn every frame for `DurationOn`, then blanked for `DurationOff`.
- **Audio / non-visual** elements (and a `nil Stimulus`) are triggered once, right after the slot's first VSYNC flip, and **do not clear the screen** — the previous frame is held for the whole slot. Place a visual just before a sound to keep it on screen while the sound plays. `TimingLog.OnsetNS` is the VSYNC reference at which the audio was triggered.

Audio elements must already be bound to the device via `PreloadDevice(exp.AudioDevice)` before the call (same precondition as `PlayStreamOfSounds`). For pure-audio streams prefer `PlayStreamOfSounds`, whose sub-frame sleep-polling gives finer timing than 60 Hz frame quantization.

Per-frame callback `PresentStreamOfStimuliFunc` adds an optional `FrameCallback` invoked once per frame, **just before each flip**. Use it to draw a persistent overlay (trial counter, frame border, fixation) on top of the stimulus, or to run real-time logic such as firing feedback the instant a response window lapses — things the plain stream functions cannot express.

```
header := stimuli.NewTextLine("3 / 40", 0, -300, control.White)
cb := func(ctx stimuli.FrameContext) error {
    _ = header.Draw(ctx.Screen)           // overlay, rendered over the stimulus
    if ctx.OnPhase && ctx.FirstFrame {    // onset of element ctx.Index
        // real-time feedback using ctx.NowNS / ctx.Events;
        // return sdl.EndLoop to stop early, any other error to abort
    }
    return nil
}
events, logs, err := stimuli.PresentStreamOfStimuliFunc(exp.Screen, elements, x, y, cb)
```

`FrameContext` fields: `Screen`, `Index`, `Frame`, `OnPhase`, `FirstFrame`, `NowNS` (pre-flip `sdl.TicksNS()`), `Elapsed`, `Events` (through the previous frame). Passing `nil` for the callback is identical to `PresentStreamOfStimuli`.

Per-stimulus onset trigger (post-flip hook) To emit a hardware TTL — or run any action — aligned to each stimulus **onset**, use `PresentStreamOfStimuliHooks`, which adds an optional `OnsetCallback` to the per-frame callback:

```
func PresentStreamOfStimuliHooks(
    screen *apparatus.Screen, elements []StreamElement, x, y float32,
    onFrame FrameCallback, onOnset OnsetCallback,
) ([]UserEvent, []TimingLog, error)

type OnsetCallback func(index int, onsetNS uint64) error
```

`onOnset` fires once per element with a stimulus onset, **immediately after** the VSYNC flip that turns it on (for a held audio element whose `DurationOn` rounds to zero frames, after the first-off-frame flip that triggers it). `index` matches the returned `TimingLog` slice and `onsetNS` equals `TimingLog[index].OnsetNS`.

This is the **post-flip** counterpart to `FrameCallback`: `FrameCallback` runs one frame earlier (pre-flip, at GPU-submission time), whereas `OnsetCallback` runs on the photon side of the frame boundary, so a trigger emitted here coincides with the logged onset instead of leading it by a frame.

```
dev := triggers.NewParallelPort("/dev/parport0") // any triggers.OutputTTLDevice
if err := dev.Open(); err != nil {
    log.Fatal(err)
}
defer dev.Close()

onset := func(index int, onsetNS uint64) error {
    return dev.Send(codes[index]) // per-stimulus code, at the displayed onset
}
```

```

}
clear := func(ctx stimuli.FrameContext) error {
    if !ctx.OnPhase && ctx.FirstFrame { // drop the lines at the start of each ISI
        return dev.AllLow()
    }
    return nil
}
events, logs, err := stimuli.PresentStreamOfStimuliHooks(exp.Screen, elements, 0, 0, cl

```

The hook runs on the timing-critical path with GC disabled, between the onset flip and the next frame, so it **must be short and non-blocking**: emit an edge (SetHigh / Send) here and clear it from a later FrameCallback or after the stream — never time.Sleep or a blocking Pulse, or a frame is missed. Elements with a nil Stimulus (pure delay / hold slots) have no onset and do not fire it. Either callback may be nil; PresentStreamOfStimuliFunc is exactly PresentStreamOfStimuliHooks(..., onFrame, nil).

For pure-audio streams the equivalent is PlayStreamOfSoundsHook(elements, onOnset), which fires onOnset once per non-nil Sound, immediately after Play(), with the same SDL-clock onsetNS (== TimingLog[index].OnsetNS). PlayStreamOfSounds is PlayStreamOfSoundsHook(elements, nil).

Audio Stimuli

```

// WAV file
snd := stimuli.NewSound(filePath)
snd.PreloadDevice(exp.AudioDevice)
snd.Play()
snd.Wait() // block until done
snd.PlaySegment(onset, offset, rampSec) // time-delimited segment with optional f

// Embedded WAV
snd := stimuli.NewSoundFromMemory(data)

// Procedural tone
tone := stimuli.NewTone(frequency, duration, volume) // duration: time.Duration; volu
tone.PreloadDevice(exp.AudioDevice)
tone.Play()

// One-shot helper (no preload needed)
stimuli.PlaySoundFromMemory(exp.AudioDevice, data)

// Embedded feedback sounds (via assets_embed)
import "github.com/chrplr/goxpyriment/assets_embed"
stimuli.PlaySoundFromMemory(exp.AudioDevice, assets_embed.BuzzerWav)
stimuli.PlaySoundFromMemory(exp.AudioDevice, assets_embed.CorrectWav)

```


Package apparatus and results

Import: `github.com/chrplr/goxpyriment/apparatus` (screen, input, gamma) Import: `github.com/chrplr/goxpyriment/results` (data file)

In normal experiments you access apparatus types through `exp.Screen`, `exp.Keyboard`, `exp.Mouse`, and `exp.Data`. Direct use of apparatus is only needed when writing custom stimulus types.

Screen

All stimulus positions use a **center-origin coordinate system**: (0, 0) is the screen center; positive Y is upward.

```
screen.CenterToSDL(x, y float32) (float32, float32) // convert to SDL top-left coords
screen.CenteredRect(pos FPoint, w, h float32) *FRect // SDL dest rect for a w×h texture
screen.MousePosition() (float32, float32) // current cursor in center coord
screen.Clear() error // fill with background color
screen.Update() error // present (VSYNC-blocks)
screen.Flip() error // alias for Update
screen.FlipTS() (uint64, error) // present + return SDL nanosecond timestamp
screen.FrameDuration() time.Duration // nominal frame duration (falls back to 1/60s)
screen.SetLogicalSize(w, h int32) error
screen.SetVSync(vsync int) error
screen.DisplayInfo() apparatus.DisplayInfo // monitor properties
screen.Destroy()
```

`FlipTS` returns `sdl.TicksNS()` captured immediately after `SDL_RenderPresent`. This timestamp is on the same nanosecond clock as SDL3 event timestamps, so `int64(event.Timestamp - onsetNS)` gives hardware-precision reaction time without any polling latency.

Keyboard

```
key, err := exp.Keyboard.Wait() // any key
key, err := exp.Keyboard.WaitKey(control.K_SPACE) // specific key
key, err := exp.Keyboard.WaitKeys(keys, timeoutMS) // first of several keys
key, rt, err := exp.Keyboard.WaitKeysRT(keys, timeoutMS) // with RT in ms from onset
key, ts, err := exp.Keyboard.GetKeyEventTS(keys, timeoutMS) // with SDL event timestamp
events, err := exp.Keyboard.GetKeyEventsTS(keys, timeoutMS) // first key + 50 ms
events, err := exp.Keyboard.CollectKeyEventsTS(keys, durationMS) // all keys during full duration
key, err := exp.Keyboard.Check() // non-blocking poll
held := exp.Keyboard.IsPressed(key) // true if key is physically held
upTS, err := exp.Keyboard.WaitKeyReleaseTS(key, timeoutMS) // wait for KEY_UP; return timestamp
exp.Keyboard.Clear() // drain SDL event queue
```

`WaitKeys` and `WaitKeysRT` return 0, nil on timeout; return `sdl.EndLoop` on ESC or window close.

IsPressed queries SDL's scancode state array — no event queue involvement. WaitKeyReleaseTS returns the KEY_UP hardware timestamp so that upTS - downTS gives nanosecond-precision press duration.

GetKeyEventsTS — two-phase collection for bilateral responses. The timeout governs how long to wait for the *first* key. After the first key arrives, the function waits an additional 50 ms for any second key before returning. This extra window is necessary because human “simultaneous” bilateral presses (e.g. both hands at once) arrive 10–50 ms apart — a non-blocking drain after the first key would miss the second key almost every time. Use GetKeyEventTS for ordinary single-key trials to avoid the 50 ms overhead.

GetKeyEventTS returns the SDL3 KeyboardEvent.Timestamp field — the nanosecond time at which the hardware key-down event was generated, on the same clock as sdl.TicksNS() and Screen.FlipTS(). This allows computing reaction time from any specific stimulus onset without manual arithmetic:

```
onset, _ := exp.ShowTS(stim1)    // nanoseconds at VSYNC flip
exp.Wait(500)
exp.ShowTS(stim2)
key, eventTS, _ := exp.Keyboard.GetKeyEventTS(responseKeys, -1)
rtToStim1 := int64(eventTS - onset) // nanoseconds
```

Mouse

```
x, y := exp.Mouse.Position()           // current position (center of screen)
btn, err := exp.Mouse.WaitPress()       // block until button pressed
btn, rt, err := exp.Mouse.WaitPressRT(timeoutMS) // with RT in ms from call site
btn, ts, err := exp.Mouse.GetPressEventTS(timeoutMS) // with SDL event timestamp (nanoseconds)
btn, err := exp.Mouse.Check()           // non-blocking poll
held := exp.Mouse.IsPressed(sdl.BUTTON_LEFT) // true if button is physically pressed
upTS, err := exp.Mouse.WaitButtonReleaseTS(btn, timeoutMS) // wait for MOUSE_BUTTON_UP;
exp.Mouse.ShowCursor(show bool) error
```

WaitPressRT mirrors Keyboard.WaitKeysRT: reaction time is measured in milliseconds from the call site. GetPressEventTS returns the SDL3 hardware event timestamp in nanoseconds, suitable for use with ShowTS. IsPressed and WaitButtonReleaseTS mirror the keyboard's IsPressed and WaitKeyReleaseTS.

GamePad

```
pads, err := apparatus.GetGamePads() // enumerate connected gamepads
defer pads[0].Close()

btn, err := pads[0].WaitPress() // block until any button pressed
btn, ts, err := pads[0].GetPressEventTS(timeoutMS) // with SDL event timestamp
```

GetPressEventTS returns the GamepadButtonEvent.Timestamp field — same nanosecond clock as Screen.FlipTS and keyboard/mouse event timestamps.

Unified Input — WaitAnyEventTS

When the response device is not fixed in advance (keyboard or mouse click), use the method on Experiment:

```
// Accept F or J key, or any mouse button, timeout after 3 s
ev, err := exp.WaitAnyEventTS(
    []control.Keycode{control.K_F, control.K_J},
    true,    // catchMouse
    3000,
)
```

Returns an apparatus.InputEvent:

```
type InputEvent struct {
    Device      apparatus.DeviceKind // DeviceKeyboard | DeviceMouse | DeviceGame
    Key         sdl.Keycode          // non-zero for keyboard events
    Button      uint32               // non-zero for mouse events
    GamepadButton sdl.GamepadButton // non-zero for gamepad events
    TimestampNS uint64                // SDL3 hardware timestamp, nanoseconds
}
```

TimestampNS is on the same clock as ShowTS, so RT computation is identical regardless of device:

```
onset, _ := exp.ShowTS(stim)
ev, _ := exp.WaitAnyEventTS(keys, true, -1)
rtNS := int64(ev.TimestampNS - onset)
```

Pass keys = nil to accept any key. Pass catchMouse = false to ignore the mouse. On timeout, returns a zero InputEvent and nil error. On ESC or quit, returns sdl.EndLoop.

ResponseDevice

ResponseDevice is a unified interface over all participant-input hardware — SDL-event-driven devices (keyboard, mouse, gamepad) **and** polled TTL devices (MEGTTLBox, DLPIO8). It is the recommended abstraction when the experiment design does not commit to a specific input modality.

```
type ResponseDevice interface {
    WaitResponse(ctx context.Context) (Response, error)
    DrainResponses(ctx context.Context) error
    Close() error
}

type Response struct {
    Source apparatus.DeviceKind // DeviceKeyboard | DeviceMouse | DeviceGamepad | Dev
    Code   uint32                     // SDL Keycode, mouse button, gamepad button, or TTL bitmask
    RT     time.Duration             // elapsed from WaitResponse call to detection
    Precise bool                       // true = hardware event timestamp; false = software poll
}
```

```
}
```

Response.Precise distinguishes two timing regimes:

Device	Precise	RT origin
Keyboard, Mouse, Gamepad	true	SDL3 hardware event timestamp (nanosecond)
MEGTTLBox, DLPIO8	false	time.Now() at poll detection ($\pm$ poll interval, ~ 5 ms)

Construct wrappers with the provided adapters:

```
// SDL-event-driven devices
rd := &apparatus.KeyboardResponseDevice{KB: exp.Keyboard}
rd := &apparatus.MouseResponseDevice{M: exp.Mouse}
rd := &apparatus.GamepadResponseDevice{GP: pad}

// Polled TTL device (MEGTTLBox, DLPIO8, or any type with ReadAll/DrainInputs)
box, _ := triggers.NewMEGTTLBox("/dev/ttyACM0")
rd := apparatus.NewTTLResponseDevice(box, 5*time.Millisecond)
```

Usage in a trial loop:

```
onset, _ := exp.ShowTS(stim)
_ = rd.DrainResponses(ctx)
resp, err := rd.WaitResponse(ctx)
// resp.RT is always valid; resp.Precise tells you whether to trust nanosecond accuracy
```

Microphone

```
// Construction
mic, err := apparatus.NewMicrophone(spec) // default recording
mic, err := apparatus.NewMicrophoneFromDevice(devID, spec) // specific device
apparatus.DefaultRecordingSpec() sdl.AudioSpec // F32LE mono 44100
devices, err := apparatus.GetRecordingDevices() []sdl.AudioDeviceID

// Fields
mic.DeviceID sdl.AudioDeviceID
mic.Stream *sdl.AudioStream
mic.Spec sdl.AudioSpec

// Methods
captureStartNS, err := mic.StartCapture() // flush buffer, resume device, return Tic
err := mic.StopCapture() // pause device
n, err := mic.ReadSamples(buf []byte) // read PCM bytes (non-blocking)
n, err := mic.Available() // bytes waiting to be read
mic.Close() // destroy stream
```

In normal experiments, open the microphone via `exp.OpenMicrophone(spec)` (see control package); `exp.Microphone` is then available throughout the experiment and closed by `exp.End()`.

VoiceKey

Detects voice onset by computing per-window RMS over F32LE PCM samples.

```
vk := apparatus.NewVoiceKey(mic, threshold float32, windowSz int) *VoiceKey
// threshold: RMS amplitude (0-1). Recommended starting value: 0.02-0.05.
// windowSz: samples per RMS window. 0 → 128 (≈ 2.9 ms at 44100 Hz).

captureStartNS, err := vk.Arm() // flush mic buffer, start capture, return TicksNS()

onsetNS, pcm, err := vk.WaitOnset(captureStartNS uint64, timeoutMS int) (uint64, []byte)
// Returns: onset timestamp (sdl.TicksNS() domain), all PCM bytes captured, nil or ErrV
```

`onsetNS` and the `imageOnsetNS` returned by `exp.ShowTS` or `screen.FlipTS` are on the same SDL3 nanosecond clock:

```
captureStartNS, _ := vk.Arm()
imageOnsetNS, _ := exp.ShowTS(picture)
onsetNS, pcm, _ := vk.WaitOnset(captureStartNS, 2000)
rtMs := int64(onsetNS - imageOnsetNS) / 1_000_000
```

Sentinel error: `apparatus.ErrVoiceTimeout`

Post-hoc helpers (for re-analysing saved WAV files):

```
sampleIdx := apparatus.ScanOnset(pcm []byte, threshold float32, windowSz int) int
// Returns the sample index of the first onset window, or -1 if not found.

onsetNS := apparatus.SampleOnsetNS(captureStartNS uint64, sampleIndex, sampleRate int)
// Converts a sample index back to a nanosecond timestamp.

names, err := apparatus.DeviceNames() map[sdl.AudioDeviceID]string
// Returns all recording devices and their human-readable names.
```

WriteWAV

```
apparatus.WriteWAV(path string, spec sdl.AudioSpec, data []byte) error
```

Saves raw PCM bytes as a standard RIFF/WAV file. Supports `AUDIO_F32LE`, `AUDIO_S16LE`, `AUDIO_S32LE`, and `AUDIO_U8`. Writes atomically via a temp file + rename.

DataFile

```
exp.Data.Add(field1, field2, ...)           // append a data row
exp.Data.AddVariableNames([]string{...})   // write column header
exp.Data.WriteDisplayInfo(info)             // append display metadata to the info fi
exp.Data.WriteParticipantInfo(info)         // append --PARTICIPANT INFO block to the
exp.Data.WriteEndTime()                    // append end time + duration to the info
```

Two files are written to ~/goxpy_data/ for each session:

File	Contents
<expname>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>.csv	Pure CSV data rows — directly importable by Excel, R, or pandas
<expname>_sub-<NNN>_date-<YYYYMMDD>-<HHMMSS>-info.txt	#-prefixed metadata: start/end time, hostname, OS, framework version, display and audio configuration, participant info

Package media

Import: `github.com/chrplr/goxpyriment/media`

Multi-movie playback with shared master-clock synchronisation, look-ahead frame conditions, and post-vsync display events for hardware- trigger alignment. Adds `MovieManager` (per-experiment owner of the clock and movie set), `Movie` (per-movie state with `PsyScope-Tahoe` semantics), and the `media/present` subpackage of OS-level vsync backends. Pure Go, .gv movies only, silent.

Full reference (precision contracts, platform support, mapping to `PsyScope` movie commands): see [MediaMovies.md](#).

Quick API summary

```
mgr := media.NewMovieManager(exp.Screen)
defer mgr.Close()

gv, _ := gvvideo.LoadGVVideo("clip.gv")
mov, _ := media.NewMovie(mgr, gv,
    media.WithTag("M1"),
    media.WithRepeat(-1),
    media.WithSize(640, 360),
    media.WithPosition(sdl.FPoint{X: -200, Y: 0}),
)
defer mov.Close()

// Look-ahead – fires from inside DrawWithoutFlip, BEFORE the frame is presented.
mov.OnAt(media.Frame(186), func(o media.Onset) { /* ... */ })
```

```

// Hardware-verified – fires from NotifyFlipped with the OS vsync timestamp.
mov.OnAtDisplay(media.Frame(186), func(o media.Onset) {
    _ = ttl.Pulse(0, 5*time.Millisecond)
})

// Display[Onset/Offset] – fires when the named tag changes visibility.
mgr.OnDisplayOnset("M1", func(o media.Onset) { /* ... */ })

// Atomic command burst (PsyScope script-line equivalent):
mgr.BeginBurst()
movieA.Pause(); movieB.Pause(); movieC.Pause()
mgr.EndBurst()

// Per-frame, mixed compositing:
exp.Screen.Clear()
mgr.DrawWithoutFlip()           // decode + render movies
fix.Present(exp.Screen, false, false) // overlay
ts, _ := exp.Screen.FlipTS()
mgr.NotifyFlipped(ts)

```

Onset precision summary

Source	Where	Meaning
LookAhead	OnAt / OnDone callbacks	sd1.TicksNS() at fire time, pre-vsync
VsyncEstimated	OnAtDisplay / Display* callbacks (fallback timer)	Post-Present FlipTS, ~vsync-period
HardwareVerified	OnAtDisplay / Display* callbacks (Stage 5 timer)	OS-measured first-pixel-visible, sub-ms

Platform status

Platform	Backend	Precision
macOS	CVDisplayLink (CoreVideo via purego)	HardwareVerified
Linux	DRM_IOCTL_WAIT_VBLANK (/dev/dri/cardN, video group)	HardwareVerified
Windows	(not yet — fallback)	VsyncEstimated
Other	(fallback)	VsyncEstimated

See [MediaMovies.md §8](#) for the Windows roadmap and [§7](#) for the trigger-synchrony contract (the wire pulse arrives ~0.5–5 ms after Onset.TimestampNS; calibrate the constant offset with a photodiode for sub-ms wire-side alignment).

Package design

Import: `github.com/chrplr/goxpyriment/design`

Data Structures

```
// Trial – one experimental trial
trial := design.NewTrial()
trial.SetFactor("condition", "congruent")
trial.GetFactor("condition") // → "congruent"
trial.Copy()                 // deep copy

// Block – a sequence of trials
block := design.NewBlock("Practice")
block.SetFactor("type", "practice")
block.AddTrial(trial, copies, randomPosition)
block.ShuffleTrials()

// Experiment design (separate from control.Experiment)
exp.Design // *design.Experiment – contains Blocks, DataVariableNames, etc.
```

Randomization

```
design.RandInt(a, b int) int           // random int in [a, b]
design.RandElement(list []T) T         // random element (generic)
design.CoinFlip(headBias float64) bool // weighted coin flip
design.RandNorm(a, b float64) float64  // truncated normal in [a, b]
design.ShuffleList(list []T)           // in-place Fisher-Yates shuffle (g
design.MakeMultipliedShuffledList(list []T, n int) []T // n shuffled copies concatenat
design.RandIntSequence(first, last int) []int // shuffled range [first, last]
```

Latin Square (Between-Subjects Counterbalancing)

```
// Registration
exp.AddBWSFactor("handedness", []interface{}{"left", "right"})

// At runtime – returns the condition for the current subject
condition := exp.GetPermutedBWSFactorCondition("handedness")

// Low-level
square, err := design.LatinSquare(elements, design.PBalancedLatinSquare)
square, err := design.LatinSquareInts(n, design.PCycledLatinSquare)
// permutation types: design.PBalancedLatinSquare, PCycledLatinSquare, PRandom
```

Package clock

Import: `github.com/chrplr/goxpyriment/clock`

```
clock.Wait(ms int)           // block for ms milliseconds
clock.GetTime() int64         // ms since package first used
clock.GetTimeNS() int64       // nanoseconds since package first used

c := clock.NewClock()         // clock relative to "now"
c.NowMillis() int64           // ms elapsed
c.NowNanos() int64            // nanoseconds elapsed
c.Now() time.Duration
c.Reset()                     // restart reference
c.SleepUntil(target time.Duration) // sleep until target offset (returns immediately)
```

Note: Prefer `exp.Wait(ms)` over `clock.Wait(ms)` in experiment code — `exp.Wait` pumps SDL events and detects ESC.

Clock domains: `GetTimeNS()` and `NowNanos()` use the Go monotonic clock (`time.Since`). SDL event timestamps from `Screen.FlipTS`, `GetKeyEventTS`, `GetPressEventTS`, and `WaitAnyEventTS` use `sdl.TicksNS()`. The two clocks have different origins and **must not be subtracted from each other** for reaction-time computation. Use the SDL-based functions exclusively for RT measurement.

Package geometry

Import: `github.com/chrplr/goxpyriment/geometry`

```
geometry.GetDistance(p1, p2 sdl.FPoint) float32
geometry.CartesianToPolar(x, y float32) (radius, angleDeg float32)
geometry.PolarToCartesian(radius, angleDeg float32) (x, y float32)
geometry.DegreeToRadian(deg float32) float64
```

Package triggers

Import: `github.com/chrplr/goxpyriment/triggers`

Provides hardware TTL signal output (EEG/MEG trigger codes) and TTL input (response pads wired over serial). Lines are **0-indexed (0-7)** throughout; bit N of a bitmask corresponds to line N.

Interfaces

```
// OutputTTLDevice – send trigger codes to recording equipment.
type OutputTTLDevice interface {
    Send(mask byte) error           // set all 8 lines from bitmask
    SetHigh(line int) error         // drive line HIGH (0-indexed)
    SetLow(line int) error          // drive line LOW (0-indexed)
    Pulse(line int, d time.Duration) error // HIGH for d, then LOW (blocks)
    AllLow() error                  // all lines LOW
    Close() error                   // AllLow + release port
}

// InputTTLDevice – read TTL inputs from response hardware.
type InputTTLDevice interface {
    ReadAll() (byte, error)          // bitmask of
    ReadLine(line int) (byte, error) // 0 or 1 (0-i
    WaitForInput(ctx context.Context) (mask byte, rt time.Duration, err error)
    DrainInputs(ctx context.Context) error
    Close() error
}
```

DLPIO8 (DLP-IO8-G, USB-CDC serial)

Implements both OutputTTLDevice and InputTTLDevice.

```
// Auto-detect (recommended): returns NullOutputTTLDevice if not found, no error.
out, portName, err := triggers.AutoDetectDLPIO8()
defer out.Close()
out.Pulse(0, 10*time.Millisecond) // 10 ms pulse on line 0

// Manual:
d, err := triggers.NewDLPIO8("/dev/ttyUSB0")
defer d.Close()
d.Send(0b00000101) // lines 0 and 2 HIGH
mask, err := d.ReadAll() // bitmask of all 8 input lines
mask, rt, err := d.WaitForInput(ctx)
```

MEGTTLBox (NeuroSpin Arduino Mega TTL box)

Implements both OutputTTLDevice and InputTTLDevice. Provides 8 TTL output lines (D30–D37) and 8 TTL input lines for a FORP response pad (D22–D29).

```
box, err := triggers.NewMEGTTLBox("/dev/ttyACM0",
    triggers.WithResetDelay(2*time.Second), // wait for Arduino boot (default 2 s)
    triggers.WithPollInterval(5*time.Millisecond),
)
defer box.Close()
```

```
// Output
box.Pulse(0, 5*time.Millisecond) // pulse line 0
box.PulseMask(0b00000011, 5*time.Millisecond) // pulse lines 0 and 1
box.Send(0b00000001) // set line 0 HIGH, all others LOW

// Input (FORP response pad)
_ = box.DrainInputs(ctx) // clear latched presses from previous trial
mask, rt, err := box.WaitForInput(ctx)
buttons := triggers.DecodeMask(mask) // []FORPButton
```

FORPButton constants (also serve as 0-indexed line numbers for bitmask operations):

```
triggers.FORPLeftBlue // 0
triggers.FORPLeftYellow // 1
triggers.FORPLeftGreen // 2
triggers.FORPLeftRed // 3
triggers.FORPRightBlue // 4
triggers.FORPRightYellow // 5
triggers.FORPRightGreen // 6
triggers.FORPRightRed // 7
```

FT232HTrigger (Adafruit FT232H, Linux)

Implements both OutputTTLDevice and InputTTLDevice. Pure-Go driver via Linux usbfs — no libftdi or D2XX installation required.

Wiring: AD0-AD7 (D-bus) → 8 TTL output lines; AC0-AC7 (C-bus) → 8 TTL input lines.

```
box, err := triggers.NewFT232H(
    triggers.WithFT232HPollInterval(5*time.Millisecond), // optional; default 5 ms
)
if err != nil { log.Fatal(err) }
defer box.Close()

box.Send(0b00000001) // AD0 HIGH (persistent)
box.Pulse(0, 5*time.Millisecond) // AD0: HIGH for 5 ms, then LOW
box.AllLow()

_ = box.DrainInputs(ctx)
mask, rt, err := box.WaitForInput(ctx)

// Auto-detect (falls back to NullOutputTTLDevice if no device found):
out, err := triggers.AutoDetectFT232H()
defer out.Close()
```

Prerequisites (Linux): - The ftdi_sio kernel module must not hold the device:
 sudo rmmod ftdi_sio - User needs rw access to /dev/bus/usb/BBB/DDD. Recommended udev rule: ACTION=="add", SUBSYSTEM=="usb", ATTRS{idVendor}=="0403", ATTRS{idProduct}=="6014", MODE="0666", GROUP="plugdev"

LinuxGPIOTrigger (Raspberry Pi and other Linux SBCs)

Implements both OutputTTLDevice and InputTTLDevice. Uses the Linux GPIO character device v2 API (kernel ≥ 5.10) — no external libraries required.

Wiring: any 8 GPIO pins for output, any other 8 for input. Pin numbers are chip-relative offsets (= BCM numbers on Raspberry Pi).

```
box, err := triggers.NewLinuxGPIOTrigger(  
    triggers.WithGPIOOutputPins([8]int{17, 27, 22, 5, 6, 13, 19, 26}),  
    triggers.WithGPIOInputPins([8]int{12, 16, 20, 21, 4, 25, 24, 23}),  
    triggers.WithGPIOChip("/dev/gpiochip0"),           // optional; this is the default  
    triggers.WithGPIOPollInterval(5*time.Millisecond), // optional; default 5 ms  
)  
if err != nil { log.Fatal(err) }  
defer box.Close()  
  
box.Send(0b00000001)           // pin 17 HIGH  
box.Pulse(0, 5*time.Millisecond) // pin 17: HIGH for 5 ms, then LOW  
  
_ = box.DrainInputs(ctx)  
mask, rt, err := box.WaitForInput(ctx)
```

Output-only and input-only configurations are both valid — omit WithGPIOOutputPins or WithGPIOInputPins as needed.

Prerequisites: - Linux kernel ≥ 5.10 - User in the gpio group: `sudo usermod -aG gpio $USER`

LabJackT4 (Modbus TCP)

Implements both OutputTTLDevice and InputTTLDevice. Pure-Go Modbus TCP driver — no LabJack SDK or system library required.

Wiring: FIO0–FIO7 → 8 TTL output lines (configured as outputs on open); EIO0–EIO7 → 8 TTL input lines (configured as inputs on open).

```
box, err := triggers.NewLabJackT4("192.168.1.100",  
    triggers.WithT4PollInterval(5*time.Millisecond), // optional; default 5 ms  
    triggers.WithT4Timeout(1*time.Second),           // optional; default 1 s  
    triggers.WithT4UnitID(1),                         // optional; default 1  
)  
if err != nil { log.Fatal(err) }  
defer box.Close()  
  
box.Send(0b00000001)           // FIO0 HIGH  
box.Pulse(0, 5*time.Millisecond) // FIO0: HIGH for 5 ms, then LOW  
  
_ = box.DrainInputs(ctx)  
mask, rt, err := box.WaitForInput(ctx)
```

The host string may include the port number ("192.168.1.100:502"); default Modbus port 502 is appended automatically if omitted.

ParallelPort (Linux LPT)

Implements OutputTTLDevice.

```
ports := triggers.AvailableParallelPorts() // scans /dev/parport0..3
pp := triggers.NewParallelPort("/dev/parport0")
if err := pp.Open(); err != nil { log.Fatal(err) }
defer pp.Close()
pp.Send(0b000000111) // lines 0,1,2 HIGH
pp.Pulse(0, 10*time.Millisecond)
status, _ := pp.ReadStatus() // Linux only: status register
```

Prerequisites: sudo modprobe ppdev; user in the lp group.

Null devices

NullOutputTTLDevice and NullInputTTLDevice are silent no-ops, safe to call without hardware. AutoDetectDLPI08 and AutoDetectFT232H return a NullOutputTTLDevice when no device is found.

Package assets_embed

Import: github.com/chrplr/goxpyriment/assets_embed

Provides embedded default assets as []byte slices, ready for FontFromMemory or PlaySoundFromMemory:

```
assets_embed.InconsolataFont []byte // default monospace TTF font
assets_embed.BuzzerWav       []byte // error/incorrect feedback sound
assets_embed.CorrectWav      []byte // correct/reward feedback sound
```

Coordinate System

All stimulus positions are in **screen-center coordinates**:

- (0, 0) = screen center
- Positive X = right; positive Y = **up** (not down)
- screen.CenterToSDL(x, y) converts to SDL's top-left origin

```
      +Y (up)
      |
-X ---+--- +X
      |
      -Y (down)
```

Typical Experiment Structure

```
package main

import (
    "log"
    "github.com/chrplr/goxpyriment/control"
    "github.com/chrplr/goxpyriment/design"
    "github.com/chrplr/goxpyriment/stimuli"
)

func main() {
    exp := control.NewExperimentFromFlags("My Experiment", control.Black, control.White)
    defer exp.End()

    exp.AddDataVariableNames([]string{"block", "trial", "condition", "key", "rt_ms"})

    // Build design
    block := design.NewBlock("main")
    for _, cond := range []string{"left", "right"} {
        t := design.NewTrial()
        t.SetFactor("condition", cond)
        block.AddTrial(t, 10, false)
    }
    block.ShuffleTrials()
    exp.AddBlock(block, 1)

    err := exp.Run(func() error {
        exp.ShowInstructions("Press F for left, J for right.\n\nPress SPACE to start.")

        for bi, blk := range exp.Design.Blocks {
            for ti, trial := range blk.Trials {
                cond := trial.GetFactor("condition").(string)

                exp.Show(stimuli.NewFixCross(20, 2, control.White))
                exp.Wait(500)

                stim := stimuli.NewTextLine(cond, 0, 0, control.White)
                exp.Show(stim)

                key, rt, err := exp.Keyboard.WaitKeysRT(
                    []control.Keycode{control.K_F, control.K_J}, 3000,
                )
                if control.IsEndLoop(err) {
                    return control.EndLoop
                }
            }
        }
    })
}
```

```

    }

    exp.Data.Add(bi, ti, cond, key, rt)
    exp.Blank(500)
  }
}
return control.EndLoop
})
if err != nil && !control.IsEndLoop(err) {
    log.Fatalf("experiment error: %v", err)
}
}

```